



SiN etching characteristics of Ar/CH₃F/O₂ plasma and dependence on SiN film density

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We evaluated the silicon nitride (SiN) etching characteristics of Ar/O₂/hydrofluorocarbon plasma. Ar/CH₃F/O₂ plasma achieved a high etching selectivity of SiN to SiO₂ by increasing the oxygen flow rate. We also evaluated the dependence of SiN etching characteristics on SiN film density. A low-density film deposited at a low temperature of 200 °C (by plasma-enhanced CVD, PECVD) showed an 8–20% lower etching rate of SiN than a high-density film deposited at a high temperature of 780 °C (by low-pressure CVD, LPCVD) when we had a low RF bias of 30 W. This PECVD film might move the competitive balance to oxidation from fluorination, reducing the SiN etching rate. However, when we have a high RF bias of more than 50 W, the SiN etching rate is 2–15% higher in the PECVD film than in the LPCVD film. The etching rate of SiN at various densities depends on the balance between oxidation and ion bombardment. © 2016 The Japan Society of Applied Physics

1. Introduction

Silicon nitride (SiN) is widely used in complementary metal–oxide–semiconductor logic devices, such as gate etching masks, various spacers, stress generators, sacrificial layers, stopper layers, and other hard masks. As the device structure becomes complicated for high performance and high integrity, we should have more precisely controlled SiN etching technology.^{1–11)} Silicon nitride is frequently used for gate spacers and gate caps, as shown in Fig. 1. Figure 1(a) shows nitride spacer etching. We have to have minimal shoulder loss, maintain critical dimensions (CDs), and have a high selectivity to underlying SiO₂ or Si. It is difficult to meet all these criteria at the same time. Figure 1(b) shows the cap removal step. Recent devices were required a low gate resistance for reducing power consumption. The replacement gate structure in which a dummy Si gate is removed after dielectric insulator formation has been used. For this purpose, we have to remove the nitride cap after filling the interlayer oxide. At that time, we have to remove the nitride cap while maintaining the height of the spacer and oxide. This is also challenging.

Different types of SiN are used to fabricate the complicated structure of ultralarge-scale integrated (ULSI) devices. SiN films with different densities have been used depending on the requirement.¹²⁾ For example, multilayered SiN spacer structures with different densities have been used for advanced ULISs to fabricate a lightly doped drain and source structure. These nitrides are deposited by chemical vapor deposition (CVD), atomic layer CVD, and other methods. For the fabrication of ULISs, we have to control the selectivity between different nitrides. In some cases, we have to have a high selectivity (Fig. 1), while other cases require the opposite. However, the dependence of etching characteristics on film density has not yet been investigated.

Researchers clarified that fluorocarbon or hydrofluorocarbon should be used in SiN etching. In the 1980s, barrel or downflow plasma was investigated in detail. These technologies exhibited a high selectivity of SiN to SiO₂ by using CF₄/N₂/O₂ and NF₃/Cl₂.¹³⁾ However, the etchings were isotropic, and we cannot apply these gas chemistries to current minute pattern etching. For the patterning of SiN,

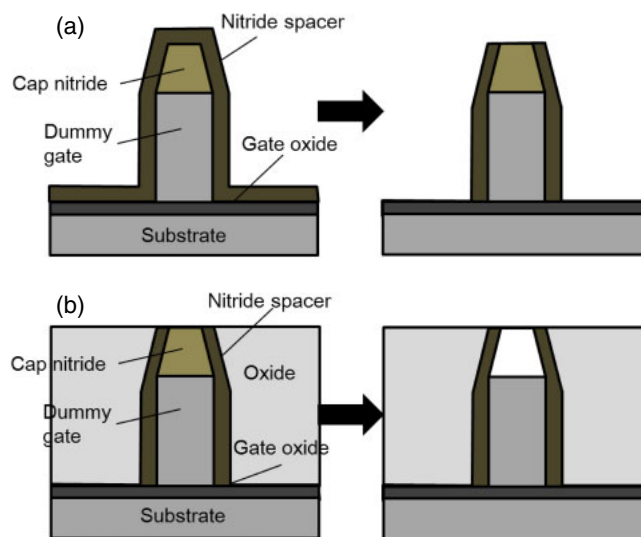


Fig. 1. (Color online) (a) Nitride spacer etching for gate and (b) cap nitride removal for replacement gate technology.

CF₄, CHF₃, CH₂F₂, and CH₃F were evaluated as etchants.^{14–19)} We generally add O₂ to hydrofluorocarbon to reduce excessive polymer deposition.

The etching mechanism was also discussed^{20–40)} for vertical and selective SiN etchings. Compared with Si and SiO₂, it is more difficult to understand the etching mechanism of SiN because SiN exhibits H density variation. At the beginning of the discussion on the etching mechanism, the fluorine (F) effect was mainly discussed.^{23–31)} Fluorine can react with both Si and N, making volatile species, such as SiF_x and NF_x. Later, some researchers pointed out that the NO radical is important for removing SiN and improving selectivity; however, the selectivity to SiO₂ was not high in that case.^{32–40)} Kim et al. showed higher etching rate and selectivity to organic materials by CH₂F₂/H₂ gas chemistry.²⁾ Loewenstein pointed out that H addition is important for increasing the SiN etching rate and improving the selectivity of SiN to SiO₂.²⁴⁾ Kataoka et al. showed etching rate enhancement by adding H₂ to CF₄/O₂.¹²⁾ We can suggest that H₂ gas addition to fluorocarbon plasma increases the

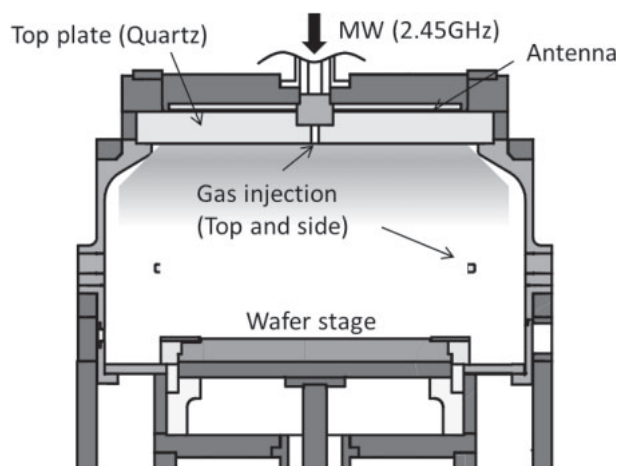


Fig. 2. Schematic of surface wave plasma tool (Tokyo Electron RLSA™ Etcher).

SiN etching rate and improves selectivity on the basis of these studies. This suggests that H in hydrofluorocarbon plasma determines the etching rate of SiN. It was also pointed out that HCN and FCN volatile byproducts contribute to enhancing SiN etching. Clarke et al. showed CN ion byproducts from SiN etching by quadrupole mass spectrometry in CF_4/O_2 plasma.⁴¹⁾ Park and coworkers and Smolinsky et al. mentioned that HCN enhanced SiN etching.^{42–44)} In the industry, CH_3F has become the standard gas for nitride etching, showing a high selectivity to oxide and Si. However, few studies have been conducted, except for Chen et al.'s work.⁴⁵⁾

For this study, we investigated SiN etching with various hydrofluorocarbons containing plasma. Comparing CF_4 and CH_2F_2 , CH_3F showed a higher etching selectivity of SiN to oxide with the addition of some amount of O_2 . We also investigated the dependence of SiN etching characteristics on the film deposition method. Surface oxidation was performed on a PECVD film, resulting in a lower etching rate; however, we observed a higher etching rate on a PECVD film when we applied a sufficiently large RF power due to a high ion energy.

2. Experimental methods

Figure 2 shows a schematic of the surface wave plasma etcher used for this experiment on a wafer with a diameter of 300 mm (Tokyo Electron RLSA™ Etcher). The properties and configuration of this tool have been described in previous studies.^{46–53)} In this etcher, plasma is generated using a 2.45 GHz microwave radiating from a radial line slot antenna. High-density plasma ($>10^{11}\text{ cm}^{-3}$) can be generated under the top plate. The plasma diffuses to the substrate, resulting in a quite low electron temperature near the wafer stage ($<1\text{ eV}$ in Ar plasma). The RF power driven by 13.56 MHz was applied to the substrate to accelerate ions to the wafers. We used 300-mm-diameter wafers in this experiment.

Three kinds of SiN having different densities were prepared by changing the substrate temperature at the depositions, as listed in Table I. In 200 and 400 °C samples, plasma-enhanced CVD (PECVD) was used with a gas mixture of SiH_4 , NH_3 , Ar, and N_2 . Low-pressure CVD (LPCVD) was used at 780 °C. The H amount, measured by secondary

Table I. Properties of SiN film used.

Deposition temperature (°C)	Hydrogen density (10^{21} cm^{-3})	Film density (g/cm^3)	Initial thickness (nm)
200	22.2	2.60	50
400	19.0	2.80	50
780	3.92	2.90	250

ion mass spectroscopy (SIMS), was higher at low-temperature deposition. Film density was measured by X-ray reflection (XRR). Kataoka et al. mentioned that their film had densities of 2.3 g/cm^3 at 300 °C and 3.1 g/cm^3 at LPCVD, which are reasonably coincident with our results.¹²⁾ We can also confirm that low-temperature deposition provides a high H amount and a low film density.

We analyzed the surface properties of the SiN film by XPS (Kratos Analytical AXIS ULTRA DLS), which has a high energy resolution after taking the wafer out from the chamber. We assigned 101.9 eV to SiN, 103.6 eV to Si–O, and 289.0 eV to CHF. In addition, we measured the optical emission from plasma by an optical emission spectroscopy system. This system has an optical resolution of 0.5 nm with a wide wavelength range from 200 to 800 nm. It has multiple optical fibers to collect optical emission 5 mm above the wafer.

3. Results and discussion

3.1 SiN etching characteristics of hydrofluorocarbon plasma

Figure 3 shows the (a) SiN etching rate, (b) SiO_2 etching rate, and (c) etching selectivity of SiN to SiO_2 as a function of O_2 flow rate at different fluorocarbon gas chemistries in Ar/ O_2 /hydrofluorocarbon plasma. The etching conditions were pressure: 40 mT, microwave (MW) power: 1.5 kW, RF power: 30 W, Ar/hydrofluorocarbon = 200/45 sccm, 15 °C, and etching time: 15 s. A positive etching rate means the removal of material, while a negative etching rate means deposition on the surface. The Ar/ CF_4 / O_2 case exhibited a high SiN etching rate without O_2 ; however, the SiN etching rate decreased as O_2 flow rate increased. The CH_2F_2 and CH_3F cases showed no SiN etching without O_2 , but the SiN etching rate increased with an O_2 flow rate of more than 15 sccm. Mele et al. pointed out that the fluorocarbon polymer might block SiN etching. A hydrofluorocarbon polymer was deposited on the SiN surface, blocking SiN etching on the basis of Mele et al.'s findings.¹⁸⁾ The CF_4 dissociation is lower than other hydrofluorocarbon dissociations because of its small cross section, causing SiN etching without O_2 . Regarding the SiO_2 etching rate [Fig. 3(b)], CF_4 showed some etching without O_2 owing to a low level of deposition, while surface deposition occurred with CH_2F_2 and CH_3F without O_2 flow. At an O_2 flow rate of 15 sccm, the surface ratio of O to C is high enough to reduce the SiO_2 etching. We then can achieve a high etching selectivity of SiN to SiO_2 at CH_2F_2 or CH_3F gas chemistry with an O_2 flow rate higher more than 15 sccm, as shown in Fig. 3(c). We do not have a clear explanation for the higher SiN etching rate in CH_3F than in CH_2F_2 . However, we can suggest several reasons. CH_3F provides a larger amount of H, which might help to increase the SiN etching rate through HCN generation. It also might help to reduce the ability of O to generate OH, increasing the SiN etching rate.

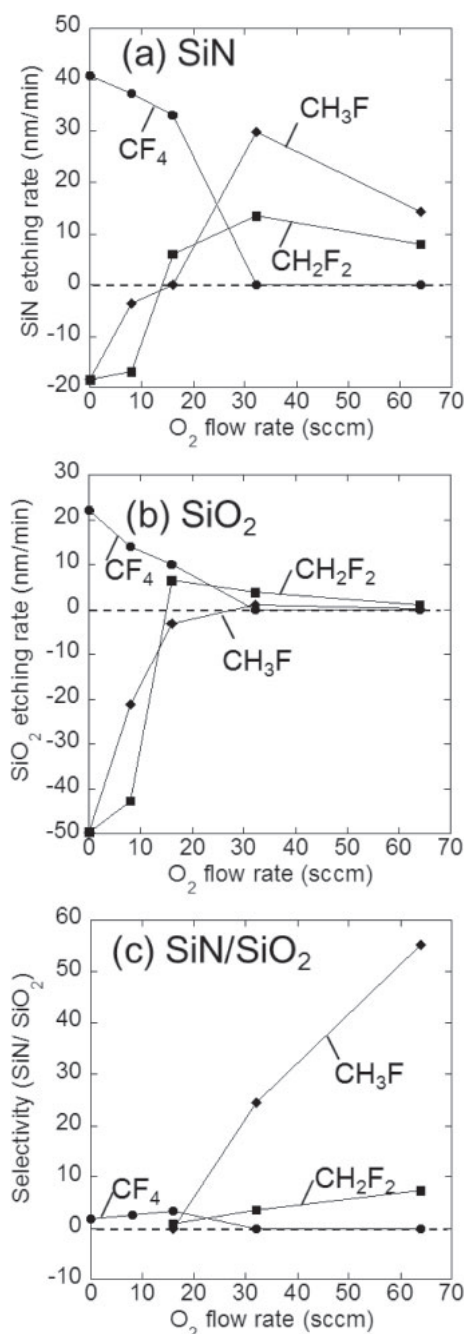


Fig. 3. Oxygen flow rate dependences of (a) nitride etching rate, (b) oxide etching rate, and (c) etching selectivity of nitride to oxide in different hydrofluorocarbon gases. Etching conditions were pressure: 40 mT, MW power: 1.5 kW, RF power: 30 W, Ar/hydrofluorocarbon = 200/45 sccm, and substrate temperature: 15 °C.

Ito et al. argued that CH₂F⁺ exhibited a higher etching yield than CF⁺, which also helps to increase the SiN etching rate in CH₃F.²²⁾ We suggest that these findings can explain why we observed a higher SiN etching rate in CH₃F.

To characterize SiN etching in hydrofluorocarbon plasma, we analyzed the plasma and surface conditions. Figure 4 shows the OES signals of hydrofluorocarbon-containing plasma. The etching conditions were pressure: 40 mT, MW power: 1.5 kW, Ar/hydrofluorocarbon/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C. There were some optical emission peaks. In the experiments with all three hydrofluorocarbon gases, the intensities of the Ar peak of 750.4 nm were within 10% variation. This means that hydro-

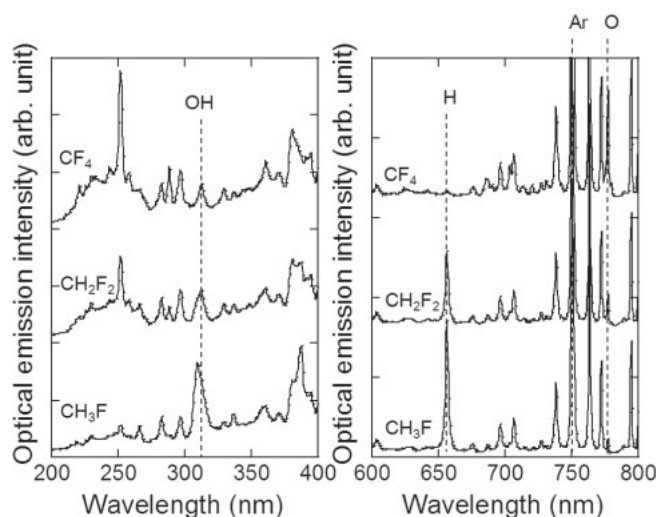


Fig. 4. Optical emission spectra of CF₄, CH₂F₂, and CH₃F gas chemistries. Etching conditions were pressure: 40 mT, MW power: 1.5 kW, Ar/hydrofluorocarbon/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C.

fluorocarbon has less impact on plasma generation under these conditions. We assigned H, OH, and O signals to 656.14, 306.4, and 777.5 nm, respectively.⁵⁴⁾ We clearly observed the H signal in CH₃F because CH₃F is dissociated to generate H radicals, which scavenge O radicals to generate OH radicals, which we observed as a strong signal in the CH₃F-containing plasma. On the other hand, the O signal was strong in CF₄ because no hydrogen radicals were formed in the plasma. This means that the SiN surface is easily oxidized and the etching rate should be reduced.

Surface conditions were evaluated by XPS. Figure 5(a) shows examples of Si 2p and C 1s XPS spectra at the plasma-exposed SiN surface. These spectra were taken at the PECVD SiN film. The plasma exposed conditions were pressure: 40 mT, MW power: 1.5 kW, Ar/CH₃F/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C. We assigned 101.9 eV to SiN and 103.6 eV to Si–O for Si 2p on the basis of a previous paper. Regarding C 1s, we had a large peak of C–C and a small peak related to hydrofluorocarbon. In this paper, we would like to call this small peak CHF. It is slightly difficult to fit the photoelectron curve by a single peak; however, we assigned 289.0 eV to CHF. Figure 5(b) shows the ratio of Si–O to Si–N intensities of Si 2p and the signal intensity of CHF on the etched SiN surface in different hydrofluorocarbon gases. The etching conditions were pressure: 40 mT, MW power: 1.5 kW, Ar/hydrofluorocarbon/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C. In the CH₂F₂ and CF₄ cases, the surface was oxidized and the Si–O signal was smaller at the CH₃F surface. The CHF remained at the CH₃F surface; however, it did not remain at the CF₄ and CH₂F₂ surfaces. Etching might proceed at the CH₃F surface because CHF remained.

Accordingly, we can suggest the following mechanism as a SiN etching mechanism on the basis of the experimental results. Figure 6 shows our estimated etching model of SiN. (Hydro)fluorocarbon radicals originate from plasma and generate a (hydro)fluorocarbon polymer. By ion bombardment, a mixing layer between the polymer and SiN is generated. In the case of CF₄ without O₂, the surface is

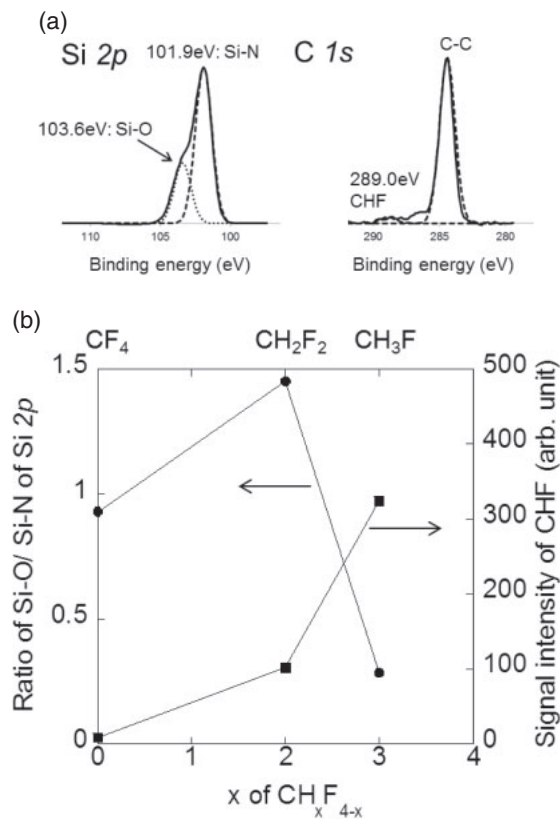


Fig. 5. (a) Si 2p and C 1s XPS spectra at plasma-exposed SiN surface. (b) Ratio of Si-O to Si-N intensities of Si 2p and signal intensity of CH_xF_y of C 1s obtained by XPS analyses. Etching conditions were pressure: 40 mT, MW power: 1.5 kW, RF power: 30 W, Ar/hydrofluorocarbon/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C.

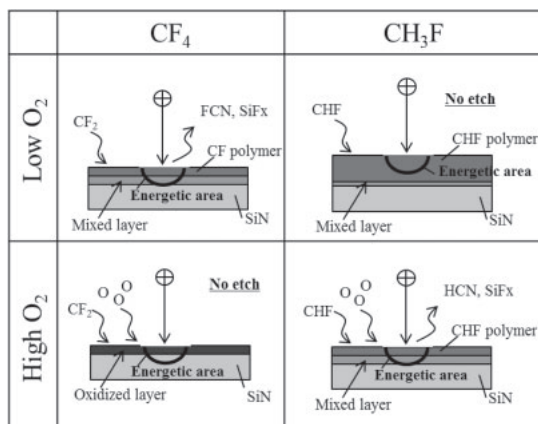


Fig. 6. Nitride etching mechanism by CF₄ and CH₃F plasmas.

covered with a stack containing a CF polymer and a mixed layer. Ion bombardment energy is provided to generate byproducts, such as FCN and SiF_x, with a high vapor pressure. However, too much CHF polymer is generated at the CH₃F surface without O₂, resulting in the absence of etching due to the absence of delivery of ion energy to the mixing layer. Oxygen addition helps to reduce this excessive CHF polymer generation and achieve SiN etching in the CH₃F case. However, the CF₄ surface with O₂ is strongly oxidized and etching is finally stopped. Because oxide etching can be reduced by adding O₂, we can conclude that CH₃F is preferable for SiN etching.

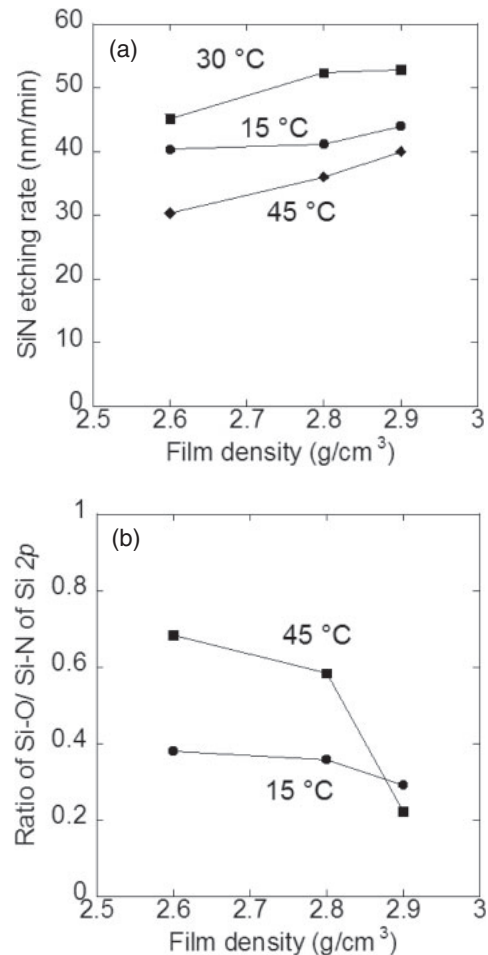


Fig. 7. (a) Nitride etching rate and (b) ratio of Si-O to Si-N intensities of Si 2p as a function of film density in nitride films at different substrate temperatures. Etching conditions were pressure: 40 mT, MW power: 1.5 kW, RF power: 30 W, and Ar/CH₃F/O₂ = 200/45/32 sccm.

3.2 Film density dependence of SiN etching characteristics

The dependence of SiN etching characteristics on SiN film density is important for highly selective SiN etching. We prepared three kinds of SiN film with different densities.

Figure 7(a) shows the SiN etching rate as a function of film density at different substrate temperatures of 15, 30, and 45 °C. The highest SiN etching rate was obtained at 30 °C. Some researchers revealed a peak of SiN etching as a function of temperature. A low substrate temperature enhances C(H)F deposition, reducing the etching rate. However, a high substrate temperature reduces the absorption efficiency of (hydro)fluorocarbons. This mechanism generates a peak of SiN etching. In Fig. 7(a), the SiN etching rates at 30 and 45 °C increased as film density increased. Figure 7(b) shows the ratio of Si-O to Si-N intensities of Si 2p XPS signals as a function of film density. The ratio was almost the same in the high-density film (2.9 g/cm³), but a large difference was observed in the low-density film (2.6 g/cm³). A low film density results in many dangling bonds and H terminations. They can be easily reacted with O radicals, generating Si-O bonds. These bonds may reduce the etching rate. This oxidation was enhanced at higher temperatures, and we observed a significant decline in etching rate at 45 °C.

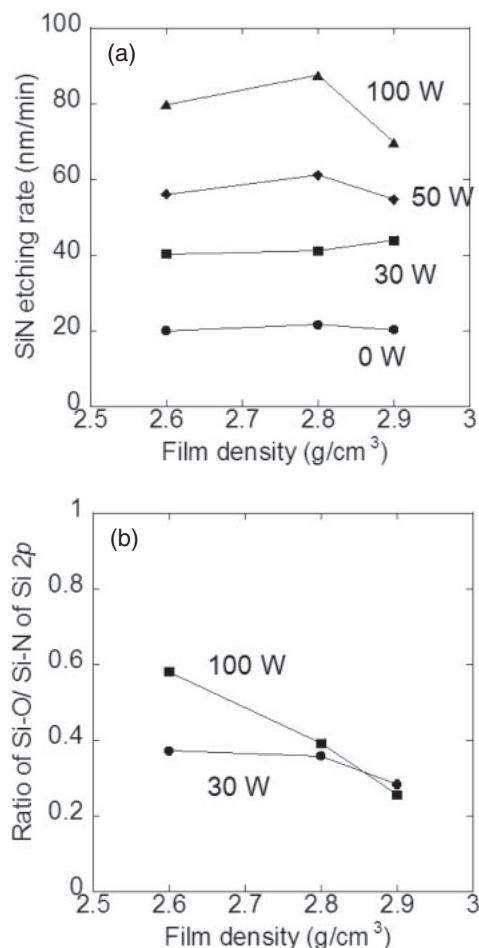


Fig. 8. (a) Nitride etching rate and (b) ratio of Si–O to Si–N intensities of Si 2p as a function of film density in nitride films at different RF powers. Etching conditions were pressure: 40 mT, MW power: 1.5 kW, Ar/CH₃F/O₂ = 200/45/32 sccm, and substrate temperature: 15 °C.

Figure 8(a) shows the SiN etching rate as a function of film density at different RF bias powers from 0 to 100 W. The etching rate increased as RF bias power increased. With this technique, RF power accelerates ions to the substrate without plasma generation. The addition of RF power increased the ion flux and energy, resulting in a high etching rate. At a low RF bias to 30 W, the etching rate was almost constant as film density increased. However, at a high RF bias above 50 W, SiN films with low densities, such as 2.6 and 2.8 g/cm³, exhibited a higher rate than the dense one (2.9 g/cm³). To explain this, we observed the exposed surface by XPS. Figure 8(b) shows the ratio of Si–O to Si–N intensities of Si 2p XPS signals as a function of film density. The Si–O intensity was increased by reducing film density at both 30 and 100 W, but at 100 W, a larger change was observed. It can be assumed that this oxidation reduces the etching rate; however, the etching rate was higher in low-density films. We believe that this can be explained by ion bombardment at the films with low densities, such as 2.8 and 2.6 g/cm³ at 100 W as compared with 2.9 g/cm³. Since low-density SiN is vulnerable, we suggest that high-energy ions can remove SiN. However, we can assume that the strong oxidation at 100 W reduces the etching rate to 2.6 g/cm³ from 2.8 g/cm³. Accordingly, it can be assumed that the etching rate is determined by the balance between ion bombardment and

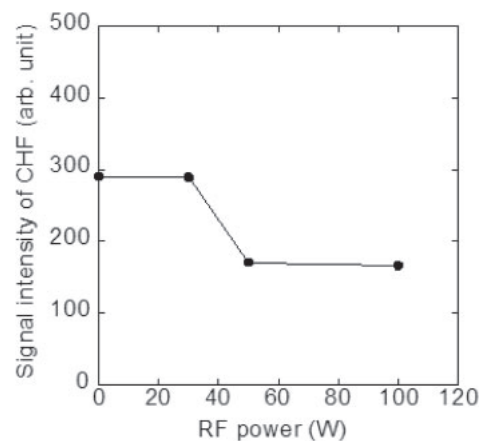


Fig. 9. Signal intensity of CHF of C 1s obtained by XPS analyses as a function of RF power. Etching conditions were the same as those in Fig. 8.

surface oxidation. Finally, we would like to discuss why we did not observe the etching rate decrease in low-density films at less than 30 W. Figure 9 shows the CHF intensity of C 1s as a function of RF bias. We can clearly see a higher intensity of CHF at less than 30 W. This means that the hydrofluorocarbon polymer was deposited at less than 30 W and the surface oxidation effect was smaller in that range. This might not cause dependence on film density at a low RF bias.

We would also like to discuss the H effect, which originates from the plasma and film. The SiN film has a certain amount of H, as shown in Table I. Assuming that the etching rate is 40 nm/min, the film deposited at 200 °C has $1.4 \times 10^{13} \text{ cm}^{-2} \text{ s}^{-1}$ H atoms. We operated the plasma, which has about $1.2 \times 10^{15} \text{ cm}^{-3}$ H molecules at 40 mT. These molecules are dissociated (typical dissociation rate: 10–80%) and reach the film surface with thermal velocity ($2.7 \times 10^5 \text{ cm s}^{-1}$ at 300 K).⁵⁵⁾ Therefore, we suggest that the H effect from the film is lower than that from the plasma.

Consequently, we can explain the dependence of the SiN etching characteristics on film density. If we have a low-density film, the surface is easily oxidized and the etching rate is lower. This is enhanced by temperature. However, if we have a high-ion energy, we can have a high etching rate even in a low-density film because such a film is vulnerable to a strong ion bombardment. This indicates that the SiN etching rate depends on the balance between ion bombardment and surface oxidation under this etching condition.

4. Conclusions

The Ar/CH₃F/O₂ plasma achieved a high etching selectivity of SiN to SiO₂ by controlling the O₂ flow. This might be because H from CH₃F scavenges the excess amount of O₂. Regarding the SiN film density, the low-density PECVD film was easily oxidized, reducing the etching rate. The low-density PECVD film also showed an increase in etching rate with high-ion energy. The SiN etching rate is determined by the balance between oxidation and ion bombardment.

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